

GATE 2010 EE, 52nd Question Analysis

Question 52

statement for linked answer questions: 52 & 53

The following Karnaugh map represents a function F .

		F			
		00	01	11	10
X	0	1	1	1	0
	1	0	0	1	0

52. A minimized form of the function F is

- (A) $F = \bar{X}Y + YZ$ (B) $F = \bar{X}\bar{Y} + YZ$ (C) $F = \bar{X}\bar{Y} + Y\bar{Z}$ (D) $F = \bar{X}\bar{Y} + \bar{Y}Z$

Question Analysis:

- From above Kmap, the output F is as follows
- $F = \bar{X}.\bar{Y} + Y.Z$
- The Truth table for above expressions is as follows

The Truth Table

X	Y	Z	$F = \bar{X}.\bar{Y} + Y.Z$
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Hardware Implementation

The above problem is implemented and tested in hardware using Arduino UNO board. Here we used 3i/ps and 7474 IC to glow LED (output F) as per truth table and verified the expression.

Required Components & Pin Connections

S.No	Component
1	Arduino Uno Board
2	Breadboard
3	7474 IC (1)
4	LEDs (1)
5	Resistors: 220 Ω (1)
6	Jumper Wires
7	USB Cable

Component	Arduino Pin
Input Z (5V or GND)	Digital 2
Input Y (5V or GND)	Digital 3
Input X (5V or GND)	Digital 4
Output F (7474 D1)	Digital 8
Output clk (7474 clk)	Digital 13
GND	GND
VCC	5V

Logic Description

- Let initialize inputs $X = 0, Y = 0, Z = 0$
- The output expressions from KMAP and is as follows
- $F = \bar{X}.\bar{Y} + Y.Z$
- Change inputs and note Down the Experimental Truth table

Code Uploading Steps

1. Create a 7474 folder and write Makefile
2. Write The code in main.c

3. Run the command "make". It will compile the code and creates .hex file
4. Copy the .hex file to ArduinoDriod folder
5. connect the Arduino UNO to mobile with OTG cable
6. Upload the hex file using "upload precompiled" option
7. Observe the ouput and verify the expression

Experimental Truth Table

X	Y	Z	F (Output)
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



Conclusion

- From Experimental Truth Table and actual truth table $F = \bar{X}.\bar{Y} + Y.Z$

- This matches option (B) from the original GATE question.
- The hardware experiment confirms the circuit's theoretical logic.